Karan Badlani

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Work Experience

MFS Investment Management | *Data Science Analyst Co-op* **Jan 2025 - Jul 2025**

- Partnered with Finance and Sales teams to develop an AI-driven forecasting and “Rising Star” analytics workflow, integrating statistical modeling and ML to provide a single source of truth for advisor quota planning and targeting high-potential segments.
- Benchmarked ARIMA, XGBoost, and Prophet models against traditional baselines, cutting total forecast error from $7.5B ’$1.65B (78%) and improving MAPE from 12.4% ’9.8%, directly influencing strategic sales decisions.
- Automated the production pipeline using Snowflake, Airflow, and Azure ML endpoints, serving rolling forecasts with cross-validation, drift monitoring, and confidence intervals for ongoing reliability and scalability.
- Collaborated with business stakeholders to translate analytical insights into targeted outreach campaigns; A/B tests across two quarters showed sales conversion gains of 13% (Q2) and 11% (Q3), informing future budget allocation.

InfoCepts | *Data Scientist* **Jul 2022 - Jul 2023**

- Delivered an Azure-based analytics platform for a city-wide bike-share system, integrating data from multiple partners and powering 3+ Power BI dashboards used by operations and leadership teams for real-time decision-making.
- Built and automated 4+ scalable data pipelines (Azure Logic Apps, Databricks) that unified partner APIs and standardized KPIs—reducing metric drift by 17% and improving data trust across departments.
- Streamlined ETL and analytics workloads with ADF, Databricks, and Azure Synapse, enabling processing of 1TB+ structured data monthly while lowering compute/storage costs by 9%.
- Deployed bike demand forecasting models (LightGBM + Prophet) that reduced prediction error (MAPE ‟35%), directly improving daily rebalancing efficiency and workforce planning.

Shri Ramdeobaba College of Engineering | *Research Scientist* **Jan 2022 - Jun 2022**

- Led development of an AWS-based air-quality analytics pipeline (S3, Glue, Redshift) centralizing sensor data for city-level monitoring, enabling stakeholders to track pollution trends across 5 districts.
- Enhanced model accuracy by 22% using advanced feature engineering (Random Forest, XGBoost), improving reliability of hotspot detection and environmental forecasting.

Academic Projects

Predictive Analytics on GCP | Statistical Modeling – Customer Churn | https://github.com/Johnny001-DS

- Built a purchase-intent model on ~1.2M sessions using Python on GCP (Cloud Storage, BigQuery, Vertex AI); engineered ~18 RFM/behavioral features (recency, session depth, cart adds, promotions, membership offers), handled class imbalance, and time-split validation showed the top 10% of scores captured ~41% of purchasers.
- Ran K-Means segmentation (k=5) on standardized RFM data; validated differences (ANOVA, chi-squared tests), profiled segments (e.g. High-Value Loyalists ~8% users driving ~27% of spend), publishing category optimization as BigQuery tables.

FiBi - Financial Analysis | Information Retrieval | https://github.com/Johnny001-DS

- Implemented an AWS-based RAG workflow using S3, Docker, Airflow and Pinecone to process 500k+ financials documents, handling ~1k test queries/day with 78% retrieval accuracy.
- Launched Flask API on AWS ECS integrating HuggingFace transformer to generate context-aware financial summaries achieving ~2s end-to-end response time, demonstrating scalable vector search and NLP integration.

Skills

- **Programming:** Python (PyTorch/TensorFlow), R, Java, SQL
- **ML & AI:** Classification, Clustering, Semi-supervised learning, Active learning Optimization NLP (transformers), Deep learning, Time-series Forecasting (ARIMA, Prophet), Neural Networks (RNN, GNN)
- **MLOps & Deployment:** Model Building & Deployment (Azure ML/GCP VertexAI), API End points (FastAPI, Flask), CI/CD (GitHub Actions), Automation (Docker, Airflow), ETL (Databricks, ADF, Alteryx)
- **Databases & Tools:** Oracle, MySQL, GCP (BigQuery), Data Visualization (Power BI, Tableau, Matplotlib/Plotly)
- **Data:** Data mining, Data modeling, Feature engineering, Large-scale structured & unstructured data (Snowflake, Azure, S3)

Education

Northeastern University **Sep 2023 - Dec 2025**
Master of Science, Data Science (GPA: 3.7/4) *Boston, MA*

- **Coursework:** Database Management, Statistics and Supervised Machine Learning, Large Language Models, Machine Learning Operations and Best Practices, Unsupervised Machine Learning models and Data Mining

Shri Ramdeobaba College of Engineering and Management **Jul 2019 - Apr 2023**
Bachelor of Engineering, Electronics & Communication (GPA: 8.6/10) *India*

Research and Publications

- Classification & Time-series forecasting with neural networks on Air Quality data with presence of outliers.